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BLACK HOLES AS POSSIBLE SOURCES OF CLOSED AND SEMICLOSED WORLDS*

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ABSTRACT

The internal structure of spacetime inside a black hole is investigated on the assumption that some limiting curvature exists. It is shown that the Schwarzschild metric inside a black hole can be attached to the de Sitter one at some spacelike junction hypersurface which represents a short transition layer. After passing the deflation stage the de Sitter space inside the black hole begins to inflate and may become a source of a new macroscopic Universe. The corresponding conformal Penrose diagrams are given. The described model may be considered as an example of "a creation of a closed or semiclosed world in laboratory". The fate of an evaporating black hole is also briefly discussed.

1. INTRODUCTION

One of the most important unresolved problems in general relativity is the problem of singularities. According to the results of Penrose and Hawking (see, for example [1]) the singularities are typical for a classical theory of gravitation. Under rather general assumptions about the properties of the matter they occur in the Universe and inside black holes. When collapse of the matter takes place then the matter density and the curvature of space–time may increase without limit near a singularity. In such circumstances the classical theory is not applicable and in particular we cannot believe in its predictions concerning the complete global structure of space–time.

On the other hand, the classical equations become unjustified at high densities of the matter and it is natural to suppose that the classical picture in this case may be changed drastically. In particular at Planck curvatures ($R \sim \frac{1}{\ell_{p\ell}^2}, \ell_{p\ell} \sim 10^{-33} cm$) quantum effects are important. If the fluctuations of metric are small then one can still use the classical notion of space–time and quantum average $< \hat{g} >$ of a metric $\hat{g}$ to describe the space–time geometry. But the quantum corrections which are large at Planck curvatures may completely change the gravitational equations. In these circumstances only the self-consistent quantum theory may answer the question about the structure of space–time at Planck curvatures. Now it is possible only to guess some general properties of such a future theory. It may happen that such guesses may become useful in the construction of a self consistent theory.

In this paper we analyze some consequences of the assumption that the gravitational equations of the complete (not known now) theory do not allow the solutions with curvature singularities. It is evident that in such a theory the energy dominance conditions used in the Penrose–Hawking–like theorems about the singularities may be violated. In what follows we assume that all invariants of space–time curvature are limited by some universal constant $\frac{1}{\ell^2}$. It is natural to use the Planck length $\ell_{p\ell} = (\hbar G/c^3)^{1/2}$ as this universal fundamental length ℓ. The following simple consideration is in favour of this hypothesis. The possibility to describe space–time by means of an average metric tensor $g = \langle g \rangle$ is related with the fact that quantum fluctuations may be considered as small fluctuations in comparison with g. If we take into account that for the fluctuations $\frac{\delta g}{\langle g \rangle} = (\frac{\ell_{p\ell}}{L})^2$ [2],[3], where L is some characteristic size of the region in consideration, then it is clear that the notion of classical geometry is applicable only after averaging of the metric over scales $L \geq \ell_{p\ell}$. The results will not depend on a particular choice of a region only if the change of $\langle g \rangle$ from one such a region to the other satisfies the condition $\frac{\Delta \langle g \rangle}{\langle g \rangle} \leq 1$. But in this case the characteristic curvature $\Delta \langle g \rangle / L^2$ is limited by the value $\frac{1}{\ell_{p\ell}^2}$. The same reasons may lead to the analogous restrictions for curvature derivatives: $|\nabla^n R| \leq \ell_{p\ell}^{-(n+2)}$. In self consistent theory such restrictions may occur as a consequence of some field equations (of course, if our hypothesis is true). The results of Ref.[4], which were obtained in the string theory, give some arguments in favour of this possibility.

If the limiting curvature exists and singularities do not arise space–time may have a regular structure. The cosmological singularities were analyzed in works [5],[6]. To avoid them under Universe collapse the hypothesis about the limiting density of matter was suggested. Namely it was supposed that at high densities the state equation of the matter is similar to the vacuum–like one*). If we neglect the anisotropy then such an assumption is equivalent to the limiting curvature hypothesis. The final stage of the Universe contraction in this case is the de Sitter–like stage and space–time allows a regular continuation. The global structure of space–time is rather trivial in this case. The problem of singularities inside black holes looks much more complicated because of the space–time in their interior is inhomogeneous and anisotropic.

The aim of this paper is to analyze the space–time structure inside a black hole in the framework of the hypothesis that the limiting curvature exists. We do not suggest any concrete choice of field equations but analyze general properties of spherically symmetric metrics which may describe the black hole interior assuming that the above hypothesis is valid. It is worthwhile noting that the space–time inside the black hole is anisotropic so that one immediately runs into the problem of anisotropy modes. That is why we require the restriction of all the curvature invariants including the square of the Weyl tensor. We also assume that at limiting curvatures the effective energy–momentum tensor takes the vacuum–like form $T_{\mu\nu} = -\Lambda g_{\mu\nu}/8\pi$. Under these assumptions certain conclusions about the structure of the interior of black hole can be reached. One of the main results of our considerations is that a closed Universe may be created inside the black hole. This possibility does not contradict the theorem proved in Ref. [7] because its assumptions (in particular the energy dominance) are violated.

The behaviour of black hole at the final stage of its evaporation also depends on the structure of space–time inside black hole. The mass of the black hole decreases, the curvature increases and if the mass is nearly equal to Planck mass, then the curvature near black hole is comparable with Planck curvature. At this stage the assumptions used in obtaining at Hawking formulae for black hole evaporation are not valid. The typical energies of emitted particles are comparable with Planck energy when the mass of black hole is of order the Planck mass and the back reaction of such particles on the gravitational field cannot be neglected. For these reasons the final stage of black hole evaporation is questionable. It is not excluded that the black holes are quantized and their remnants (that is black holes with masses $m \sim m_{p\ell}$) are stable [8]. In this paper we use the limiting curvature hypothesis to make some remarks concerning the fate of the evaporating black hole.

*) The problems which may arise when one considers a rather general class of state equations were discussed in Ref. [18].

The paper is organized in the following manner: In Sec. 2 the space–time structure inside the eternal black hole is considered. To describe the transition region between the Schwarzschild solution and the de Sitter one we use the "massive shell" thin formalism first elaborated by Israel [9]. In Sec. 3 we investigate a more realistic situation when the black hole is the result of gravitational collapse. Sec 4 is devoted to the discussion of the final state problem for evaporating black holes and in Sec. 5 the "future evolution" of black hole interior (in particular, the possibilities of creation of closed and semiclosed Universes) is considered. All necessary formulae for thin shells are given in the Appendices. In this work we will use the units $\hbar = c = G = 1$ and sign conventions as in book [3].

2. ETERNAL BLACK HOLE

Here we will consider only spherically–symmetric black holes with masses exceeding Planck mass. This simple example is very useful for the understanding of the general situation.

First, the structure of the space–time outside collapsing matter will be investigated. For that purpose the ideal model of "eternal black hole" will be exploited. Let the metric of black hole be

$$ds^2 = -(1-2m/r)dt^2 + (1-2m/r)^{-1}dr^2 + r^2 d\Omega^2$$

$$d\Omega^2 \equiv d\theta^2 + \sin^2\theta d\varphi^2. \tag{2.1}$$

We shall use the invariant

$$\mathcal{R}^2 \equiv R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta} \equiv C_{\alpha\beta\gamma\delta}C^{\alpha\beta\gamma\delta} + 2R_{\alpha\beta}R^{\alpha\beta} - \frac{1}{3}R^2 \tag{2.2}$$

to characterize the curvature of space–time. Here $C_{\alpha\beta\gamma\delta}$ is the Weyl tensor. In such a case the hypothesis about the existence of the limiting curvature takes the form

$$\mathcal{R}^2 \leq \frac{\alpha}{\ell^4}, \tag{2.3}$$

where ℓ is some characteristic (Planckian) length and α is a dimensionless constant. For the metric (2.1) we have

$$\mathcal{R}^2 = \frac{48m^2}{r^6} \tag{2.4}$$

and correspondingly, the radius r_0 at which the curvature (2.4) gets maximum value (2.3) is equal to

$$r_0 = \left(\frac{12}{\alpha}\right)^{1/6}\left(\frac{2m}{\ell}\right)^{1/3}\ell. \tag{2.5}$$

Thus we will use the metric (2.1) (more exactly its Kruskal extension) up to radius r_0 ($\infty > r > r_0$). It is worth noting, that for the large black holes ($m >> m_{p\ell}$) the radius r_o is much larger than $\ell (r_0 >> \ell)$ and therefore the approximation of background field for the metric $g (g = \langle g \rangle)$ to describe the space–time structure at these scales is justified.

Since, according to the limiting curvature hypothesis the value of $\mathcal{R}^2$ cannot increase at $r < r_0$, the metric of space–time at $r < r_0$ must differ from the metric (2.1). This leads to some modification of Einstein equations. The corresponding equations can be written in the form

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi T_{\mu\nu} \tag{2.6}$$

where $T_{\mu\nu}$ is some nonvanishing unknown tensor. In other words, the Ricci tensor $R_{\mu\nu}$ is not equal to zero at $r < r_0$.

We suppose (our second main hypothesis) that the tensor $R_{\mu\nu}$ gets the form

$$R_{\mu\nu} = \Lambda g_{\mu\nu} \tag{2.7}$$

when the curvature invariants reach the Planck values. For the spherically–symmetric space equations (2.7) have only the de Sitter solution

$$ds^2 = -\frac{dr^2}{(r/\ell)^2 - 1} + \left[\left(\frac{r}{\ell}\right)^2 - 1\right]dt^2 + r^2 d\Omega^2 \tag{2.8}$$

where $\ell^2 = 3/\Lambda$. If the length $\ell = (3/\Lambda)^{1/2}$ is identified with the length ℓ from formula (2.3), then the value of α is equal to 24*).

Let de Sitter space be at $r < r_1$ ($\ell << r_1 \leq r_0$). For $r_1 >> \ell$ the coordinate t in (2.8) is space–like at $r_1 > r > \ell$.

The most general spherically symmetric metric in the transition region $r_1 > r > r_0$ takes the form

$$\begin{aligned} ds^2 &= -\frac{dr^2}{g(r)} + f(r)dt^2 + r^2 d\Omega^2 = \\ &= -d\tau^2 + F(\tau)dt^2 + r^2(\tau)d\Omega^2 \end{aligned} \tag{2.9}$$

The choice of signs in (2.9) ($g > 0, f > 0, F > 0$) is fixed if we demand that the sign of the invariant $\nabla r \cdot \nabla r$ in the transition region is the same as it has at $r > r_0$ and at $r < r_1$. The coordinate τ is connected to r in the following way:

$$d\tau = -\frac{dr}{\sqrt{g(r)}} \tag{2.10}$$

*) If we use another curvature squared invariant instead of $\mathcal{R}^2$, then the quantity α in (2.3) may take another value for the same metric (2.8).

and

$$F(\tau) \equiv f(r(\tau)). \tag{2.11}$$

For the metric (2.9) the invariant $C^2 \equiv C_{\alpha\beta\gamma\delta}C^{\alpha\beta\gamma\delta}$ is positive. On the other hand, the tensor R^ν_μ is diagonal and $R^2_2 = R^3_3$

$$R^\mu_\nu \equiv \mathrm{diag}(R^0_0, R^1_1, R^2_2, R^2_2). \tag{2.12}$$

Hence it follows that

$$4R^\nu_\mu R^\mu_\nu - R^2 = (2R^2_2 - R^0_0 - R^1_1)^2 + 2(R^0_0 - R^1_1)^2 \geq 0. \tag{2.13}$$

Therefore for the spherically-symmetric metric of general form (2.9) the invariant $\mathcal{R}^2$ is always nonnegative. Moreover, in this case the restriction for the value of $\mathcal{R}^2$ implies the restrictions for the values of each of the invariants $C^2, R_{\alpha\beta}R^{\alpha\beta}$ and R^2. The condition $4R^\alpha_\beta R^\beta_\alpha = R^2$ takes place if and only if the relation (2.7) is valid. For this reason the assumption about the de Sitter-like stage at critical curvatures is equivalent to the statement that the invariants $R^\mu_\nu R^\nu_\mu$ and R^2 increase in the transition region so that finally the value of R^2 reaches the limiting value $4R^\mu_\nu R^\nu_\mu$. If the above mentioned assumptions are true, then the de Sitter metric (2.8) ($r < r_1$) may be attached continuously to the Schwarzschild metric (2.1) ($r > r_0$) through the transition region ($r_1 < r < r_0$). In the general case, the global space–time structure inside the black hole may depend on the details of the transition region. Let us show that for the short duration of the transition layer ($\Delta\tau << r_0$) this dependence is insignificant and to attach the metric (2.1) to (2.8) it is only necessary to know the global characteristics of this layer.

Before the transition region ($\tau < \tau_0$) we have $T_{\mu\nu} = 0$ and after it ($\tau > \tau_1$), $T_{\mu\nu} = -(\frac{\Lambda}{8\pi})g_{\mu\nu}$. Let us introduce the notation

$$S^\mu_\nu = \int_{\tau_0}^{\tau_1} T^\mu_\nu d\tau. \tag{2.14}$$

To sew the Schwarzschild and de Sitter metrics up we can use the "thin massive shells" method developed by Israel [9]. (This method is discussed in detail in Ref. [3]; see also Appendix 1). It is worth noting that one of the main features of our problem is using space-like hypersurface for junction of the metrics, meanwhile time-like and null hypersurfaces are usually considered (see e.g. [10]).

Let $r = r_0$ be the equation of thin shell (according to the above-mentioned assumption, r_0 and r_1 are nearly the same). Then the conditions which are necessary for junction of the different metrics on the hypersurface $\sum$ take the following forms:

(i) the three geometrics induced by the de Sitter and Schwarzschild metrics on $\sum$ must be identical.

(ii) the jump of the extrinsic curvature $[K^m_n] \equiv K^{(+)m}_n - K^{(-)m}_n$ is defined by S^m_n:

$$S^m_n = -\frac{1}{8\pi}([K^m_n] - \delta^m_n[K]) \tag{2.15}$$

where $K = K^m_m$ and $m, n = 1, 2, 3$.

In the case under consideration these conditions lead to the following relations (see Appendix 2)

$$S^t_t = \frac{\lambda}{4\pi}, S^\theta_\theta = S^\varphi_\varphi = \frac{\mathcal{H} + \lambda}{8\pi}, S^r_r = 0 \tag{2.16}$$

where

$$\mathcal{H} \equiv [K^t_t] = \{\frac{r_0}{\ell^2}[(\frac{r_0}{\ell})^2 - 1]^{-1/2} + \frac{m}{r_0^2}(\frac{2m}{r_0} - 1)^{-1/2}\}$$

$$\lambda \equiv [K^\theta_\theta] = -\frac{1}{r_0}\{(\frac{2m}{r_0} - 1)^{1/2} - [(\frac{r_0}{\ell})^2 - 1]^{1/2}\}, \tag{2.17}$$

Using the definition (2.5) for r_0 and taking into account the condition $r_0 >> \ell$ it is easy to find that

$$\mathcal{H} \simeq \frac{1}{\ell}(\frac{\beta}{2} + 1), \quad \lambda \simeq -\frac{1}{\ell}(\beta - 1) \tag{2.18}$$

where $\beta \equiv (\alpha/12)^{1/4}$.

In particular, for $\alpha = 24$ we have $S^t_t < 0$ and $S^\theta_\theta = S^\varphi_\varphi > 0$. It means that for a particular choice of the null vector ℓ^μ the value of $T_{\mu\nu}\ell^\mu\ell^\nu$ may be negative in the transition region and the conditions which are necessary for singularities (in accordance with Penrose theorem) are not fulfilled here. This statement is rather general since $S^r_r = 0$ for the shell and consequently the energy density must change sign in the transition region.

It is important that the large parameter m/ℓ does not enter Eq. (2.18) so that $S^n_m \sim \ell^{-1}$. Since in transition region the tensor T^n_m must be comparable with Planck tensor $T^\mu_\nu \sim \ell^{-2}$, then $\Delta\tau \sim S^n_m/T^n_m \sim \ell$ and hence there is no contradiction with the assumption about the short duration of the transition regime ($\Delta\tau \equiv \tau_1 - \tau_0 \sim \ell$).

It is worth noting that the above expressions have the most simple form for $\alpha = 12$. Namely we have $\beta = 1$ and

$$\lambda = 0, \mathcal{H} = \frac{3}{2\ell}\frac{\gamma}{\sqrt{\gamma^2 - 1}}, r_0 = \gamma\ell \tag{2.19}$$

where $\gamma = (2m/\ell)^{1/3}$. Although this choice is rather natural, nevertheless, we cannot exclude other values for α. That is why in what follows we shall not specify the value of α.

Let us now discuss the space–time structure of eternal black hole with the de Sitter like internal region.

In Figs. 1 and 2 the standard Penrose conformal diagrams for de Sitter space and for the eternal black hole are shown. (The freedom in the choice of coordinates for the black hole metric was used to represent its diagram in a more convenient form.) Then in Fig. 3 the full Penrose conformal diagram for eternal black hole with de Sitter–like internal region is represented. It is obtained from the diagrams in Figs. 1 and 2 in which the regions $r > r_0$ (for de Sitter space) and $r < r_0$ (for black hole) are removed with the subsequent gluing these diagrams together along the hypersurface $\sum$ $(r = r_0)$. The topology of $\sum$ is $S^2 \times R^1$. This hypersurface is infinite (in the t–direction) tube of the radius r_0 (Fig. 4).

3. BLACK HOLE AS A RESULT OF MATTER COLLAPSE

In a more realistic situation when the black hole is formed by the collapsing matter, the space–time structure differs from that shown in Fig. 3. The difference is due to the matter presence. For a sake of simplicity, let us consider the spherically–symmetric homogeneous dust cloud. Then the metric inside matter (up to the Planck curvature) will be

$$ds^2 = a^2(\eta_-)[-d\eta_-^2 + d\chi^2 + \sin^2\chi d\Omega^2], \tag{3.1}$$

where

$$a(\eta_-) = a_0(1 - \cos\eta_-) \tag{3.2}$$

and $0 \le \chi \le \chi_0 < \pi/2$. The parameter $\eta_- = \pi$ corresponds to the maximum expansion of dust cloud, when $a(\pi) = 2a_0$. The internal mass of the cloud M is constant and it is equal to

$$M = \frac{3}{2}\, a_0(\chi_0 - \sin\chi_0\cos\chi_0). \tag{3.3}$$

The radius of the cloud evolves according to the law

$$r = a(\eta_-)\sin\chi_0. \tag{3.4}$$

The external mass of the cloud which differs from M due to the gravitational self–interaction is equal to

$$m = a_0\sin^3\chi_0. \tag{3.5}$$

After the boundary of the cloud intersects the surface $r = 2m$, it enters the black hole.

Before considering this case we briefly discuss the evolution of the complete closed Friedmann Universe. Its metric takes the form (3.1) and (3.2) with $0 \le \chi \le \pi$.

For the curvature invariant $\mathcal{R}^2$ we have

$$\mathcal{R}^2 = 60\frac{a_0^2}{a^2(\eta_-)} \tag{3.6}$$

and this invariant gets the maximum value (2.3) at

$$a(\eta_-) = \tilde{a} \equiv \left(\frac{60}{\alpha}\right)^{1/6}\left(\frac{a_0}{\ell}\right)^{\frac{1}{3}}\ell. \tag{3.7}$$

Since $a_0 >> \ell$ then $\tilde{a} >> \ell$. Let η_0 be the time η_- when the scale factor a reaches the value (3.7).

In accordance with the "limiting curvature hypothesis" we assume that (after some transition region) the curvature of space–time reaches its limiting value and the metric coincides with the de Sitter one:

$$ds^2 = a^2(\eta_+)[-d\eta_+^2 + d\chi^2 + \sin^2\chi d\Omega^2] \tag{3.8}$$

where

$$a(\eta_+) = \ell/\sin\eta_+ \tag{3.9}$$

Assuming that the transition layer is short and using the massive thin shells approach one can find the conditions on the junction hypersurface $\sum$ (see Appendix 2)

$$a_+ = a_- \quad \text{on} \quad \sum,$$

$$S_n^m = -\frac{\lambda}{4\pi}\;\delta_n^m, \tag{3.10}$$

where

$$\lambda = \left[\frac{1}{a_+}\frac{da_+}{d\tau} - \frac{1}{a_-}\frac{da_-}{d\tau}\right]_{r_0} = \frac{1}{\tilde{a}}\left[\sqrt{2\frac{a_0}{\tilde{a}} - 1} - \sqrt{\left(\frac{\tilde{a}}{\ell}\right)^2 - 1}\right], \tag{3.11}$$

$a_+ \equiv a(\eta_+)$, $a_- \equiv a(\eta_-)$, $d\tau = ad\eta$ and $r = r_0$ is the equation of the junction hypersurface $\sum$. When $\tilde{a} = \ell(2a_0/\ell)^{1/3}$, then $\lambda = 0$. This happens at $\alpha = 15$. In this case the solutions are smoothly sewed together withoug any "shells". However the curvature invariant $\mathcal{R}^2$ jumps on $\sum$ from $15/\ell^4$ to $24/\ell^4$, meanwhile the energy density changes continuously. In a more general case, the transition layer is needed. For $a_0 >> \ell$ we have

$$\lambda \simeq \frac{1}{\ell}\left[\left(\frac{\alpha}{15}\right)^{1/4} - 1\right] \tag{3.12}$$

and, again, the assumption about thin transition region $(\Delta\tau << \tilde{a})$ is non–contradictory. When we have only a part of closed Friedmann world $(0 \le \chi \le \chi_0 < \pi/2)$, then the condition (3.7) may be rewritten in the form

$$r_0 = \left(\frac{15}{\alpha}\right)^{1/6}\left(\frac{2m}{\ell}\right)\ell, \tag{3.13}$$

where m is the mass of the dust cloud (3.5) and r_0 is the radius of its boundary at the time when the limiting curvature (2.3) is reached. It is worth noting that the value of r_0 in (3.13) is nearly the same as r_0 given by formula (2.5).

The conformal diagram for the black hole which was formed by collapsing dust matter is shown in Fig. 5. There exist three different regions denoted as K, D and F in which the metrics coincide with Schwarzschild–Kruskal, de Sitter and Friedmann metrics, correspondingly. In the general case at the boundaries FD and KD there are transition layers. The boundary KF between the Friedmann world and the Schwarzschild–Kruskal space is regular without any "shells". The corresponding jump conditions at FD lead to the relations (3.4) and (3.5).

4. FINAL STATE OF EVAPORATING BLACK HOLE

In this Section we will give some speculations about the final stage of evaporating black hole under the assumption of "limiting curvature" existence. According to the results of Hawking the black hole must lose its mass owing to quantum evaporation [11]. To describe the black hole with variable mass we will use the Vaidya metric [12]

$$ds^2 = f dv^2 + 2dvdr + r^2 d\Omega^2 \tag{4.1}$$

$$f = \frac{2m(v)}{r} - 1 \tag{4.2}$$

where v is the advanced time coordinate (at infinity $f = -1$ and $v = t + r$). This metric is the solution of Einstein equations for the following energy momentum tensor [12].

$$T_{\mu\nu} = \frac{1}{4\pi r^2}\frac{dm}{dv}v_{,\mu}v_{,\nu}. \tag{4.3}$$

Such an energy–momentum tensor describes a spherically–symmetric flow of radiation onto the black hole. When $dm/dv < 0$ then the energy density of this flow is negative. Of course, the solution (4.1)–(4.2) is not an exact solution for a space–time of an evaporating black hole. Nevertheless, it may be used as a model metric to describe the decrease of the mass of evaporating black hole [13].

Coefficient f in (4.1) has the following invariant sense

$$f = -\nabla r \cdot \nabla r \tag{4.4}$$

The surface where $f = 0$ and hence ∇r is a light–like vector is called an apparent horizon. For a static black hole the apparent horizon coincides with the event horizon.

When the black hole mass is variable then the space–time structure may differ from that described in the previous Section. To analyze it let us attach the Vaidya metric to the de Sitter metric. As earlier the junction hypersurface is defined by the condition (2.3). The invariant $\mathcal{R}^2$ for the Vaidya metric takes the following simple form

$$\mathcal{R}^2 = \frac{48m^2(v)}{r^6} \tag{4.5}$$

The equation for hypersurface $\sum$ is

$$r = r_0(v) \equiv \left(\frac{12}{\alpha}\right)^{1/6}\left(\frac{2m(v)}{\ell}\right)^{1/3}\ell \tag{4.6}$$

The main difference between (2.5) and (4.6) is the dependence of the mass m and the radius r_0 on advanced time v in (4.6). The corresponding jump conditions at $\sum$ are given in Appendix 2. The apparent horizon and the hypersurface $\sum$ are schematically shown in Fig. 6. These hypersurfaces intersect each other at

$$r = \ell/\beta \tag{4.7}$$

where $\beta = (\frac{\alpha}{12})^{1/4}$. For $r > \ell/\beta$, the apparent horizon is outside the shell and for $r < \frac{\ell}{\beta}$ it is absent.

Let us suppose that after the mass of the black hole reaches the value

$$m_{min} = \sigma\ell \tag{4.8}$$

its further evaporation stops. First of all, it is worth noting that not all values for σ are permissible. The Vaidya and de Sitter metrics must induce the same intrinsic geometry on $\sum$. Because of that the hypersurface $\sum$ must be either space–like or time–like for both metrics simultaneously. But this condition does not hold for some values of the parameter σ. For $\beta > 1$ the range $1/2\beta < \sigma < \beta^2/2$ is forbidden; for $\beta < 1$ the range $1/2 < \sigma < 1/2\beta$ is forbidden and for $\beta = 1$ such a range is absent. If $\beta \geq 1$ and $\sigma > \beta^2/2$ (or $\beta \leq 1$ and $\sigma > 1/2\beta$), then the black hole with mass $m_{min} > \beta^2\ell/2$ ($\beta \geq 1$) or $m_{min} > \ell/2\beta$ ($\beta \leq 1$) is formed within which the "closed world exists". The conformal diagrams for these cases are analogous to that shown in Fig. 5. For both cases ($\beta \geq 1$ and $\beta \leq 1$) $m_{min} > \ell/2$.

If $\beta \geq 1$ and $\sigma < 1/2\beta$ (or $\beta \leq 1$ and $\sigma < 1/2$) then some specific semiclosed world appears as the result of black hole evaporation and $m_{min} < \ell/2$ (see Figs. 7). Since the external parameters (mass, radius) of such worlds are smaller than Planck ones, then the quantum fluctuations can disconnect the formed world from our world. In this case the black hole disappears and the formed world is closed.

If we fix $\beta = 1$ and consider two limits for σ ($\sigma \to \frac{1}{2} + 0, \sigma \to \frac{1}{2} - 0$), then we find that two different diagrams are possible: one (for $\sigma \to \frac{1}{2} + 0$) is analogous to the diagram given in Fig. 5 and the other (for $\sigma \to \frac{1}{2} - 0$) is represented in Fig. 8. In the first case, the black hole achieves the limiting mass ($m_{min} = \frac{\ell}{2}$) for an infinite time v and in second case this time is finite.

5. "THE FUTURE" OF BLACK HOLE

The de Sitter metric is often used in cosmology as the simplest possible solution which can describe the stage of inflation in a very early Universe. It should be stressed that in our case the de Sitter space is necessarily created at the stage of its deflation while in cosmology one usually is interested in the inflationary stage. In our case the existence of the phase of deflation may create some problems. One of the questions which may arise is: does the deflating de Sitter Universe inside the black hole "remember" any "quantum numbers" which characterize the collapsing matter? If the deflating Universe does not possess such a memory then there is a possibility that at the end of deflation this Universe can just disappear in the process of quantum annihilation. In what follows we consider other possibilities.

First of all it should be noted that usually the de Sitter space considered as a model in cosmology is unstable. Its type of instability, generally speaking, depends on the nature of Λ-term. Moreover the result of the decay of the de Sitter space due to this instability is defined by the effective hypersurface at which this decay occurs. Since in our case we have no concrete physical model which describes how this vacuum-like energy-momentum tensor arises at the limiting curvatures we can only guess about the fate of the de Sitter world in the black hole interior. In these speculations it is natural to suppose (and we shall do so) that the properties of this interior de Sitter space are similar to those in the usual inflationary models [14], [15]. Having this in mind we discuss now a possible fate of this de Sitter space.

At first we consider the case of the eternal black hole. If the de Sitter space "remembers" all the "quantum numbers" then there is a possibility that after inflation a new "eternal" black hole will arise in future possessing the same parameters as the original one (Fig. 9). The corresponding solution will possess the reversal invariance with respect to the time symmetry moment. It is also possible the situation (shown in Fig. 10) when the development of the interior of the black hole will produce the white hole in a new asymptotically flat Universe which lies in absolute future with respect to original asymptotically flat space. Such a process of the white hole creation may happen only if the de Sitter vacuum decays on the hypersurface r = const in coordinate system where the metric of the de Sitter Universe takes the form (2.8).

It is more probable that all (or almost all) the information about the collapsing matter would be lost at the stage of deflation. In this case there is the possibility that the decay of the de Sitter world will create the Friedmann Universe. The result of such a decay depends on some "effective" hypersurface on which this decay occurs. This hypersurface is defined by the physical process which leads to the de Sitter vacuum instability. In well known examples [14],[15] this instability is due either to the quantum metric fluctuations [16] or classical evolution of the field which gives Λ-like state equation [14]. In both cases the decay of the de Sitter world occurs on the hypersurface of constant time defined in expanding/contracting the Friedmann coordinate system. Let us suppose that it also

happens in our case. The de Sitter metric can be represented in the form

$$ds^2 = \frac{\ell^2}{\sin^2 \eta_+}(-d\eta_+^2 + d\chi^2 + \sin^2\chi\, d\Omega^2) \tag{5.1}$$

which corresponds to the closed Friedmann Universe ($k = +1$). On the other hand the metric for the closed Universe filled by the matter with an arbitrary state equation takes the form

$$ds^2 = a^2(\eta_+)(-d\eta_+^2 + d\chi^2 + \sin^2\chi d\Omega^2) \tag{5.2}$$

The metric (5.2) can be easily attached to (5.1) since they have the same functional forms. If we choose the hypersurface η_+ = const in such a manner that the extrinsic curvature does not jump then these two metrics can be sewed together withoug any "shells".

In principle the decay may happen at the deflation phase ($\eta_c < -\frac{\pi}{2}$). But it is also possible that the instability at this stage would be suppressed by the limiting curvature existence. So we expect that the decay of the de Sitter space is more probabile at the stage of inflation. In this case a new closed macroscopic Universe may arise in the interior of a black hole. For an observer living in the initial asymptotically flat space this Universe would lie in the absolute future. It should be also mentioned the possibility of the flat Friedmann Universe "creation". It happens when the de Sitter Universe "decays" on the hypersurface defined by the equation η_0 = const in flat Friedmann coordinates

$$ds^2 = \frac{1}{H^2\eta_0^2}(-d\eta_0^2 + d\chi^2 + \chi^2 d\Omega^2).$$

The corresponding conformal diagram is shown in Fig.12. Although this possibility or the possibility of the open ($k = -1$) Friedmann Universe creation are not excluded from the point of view of compatibility of the corresponding conformal diagrams, nevertheless these possibilities look rather questionable.

It was shown earlier that the process of the quantum evaporation does not change essentially the main features of the de Sitter interior. That is why we may expect that the above consideration concerning the fate of this interior is also applicable in this case.

It should be also stressed that the considered model (see e.g. Fig. 11) may be interpreted as the creation of the Universe in a laboratory via a black hole which may be formed by contraction of matter up to high density. This conclusion contradicts the results of Ref. [7] and the reason is the following. In our case the conditions of the existence of the global Cauchy surface and $T_{\mu\nu}\ell^\mu\ell^\nu \geq 0$ which are used in [7] may be violated.

6. CONCLUSIONS

In this paper we made some remarks concerning the possible internal structure of a black hole in the framework of the hypothesis about the limiting curvature existence.

Under this assumption it was shown that the closed or semiclosed macroscopic Universe may be created as a result of the gravitational collapse. There is also a possibility that the remnant of a black hole in our Universe may become a white hole in a new Universe. It should be stressed that the conformal diagrams describing these processes (see e.g. Figs. 3 and 9) to some extent resemble the maximal analytical continuation either the Reissner–Nordström or Kerr metric. One of the main differences is that in our case one may expect the stability of the Cauchy horizons, while the Cauchy horizons in the Reissner-Nordström or Kerr spacetimes are shown to be unstable [20] (see also [21]). This instability is related with an infinite blue–shift of signals sent into a black hole from external space and registered by an observer crossing the Cauchy horizon. This effect combined with the classical or quantum radiation falling down into a black hole will result in the divergence of T_μ^ν near the Cauchy horizon and its instability. In our case if only the hypothesis about the limiting curvature is valid the backreation of the matter does not allow T_μ^ν (and hence $\mathcal{R}^2$) to grow without limit and after the curvature reaches its limit we would have the de Sitter space. Hence we may expect that in our case there is no instability. In this sense the possibility (which was discussed earlier in connection with the Reissner–Nordström or Kerr space–time) "to travel" from our Universe into a new one which is in absolute future with respect to us may still become open.

Acknowledgments

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APPENDIX 1

Massive thin shells in General Relativity

Three dimensional hypersurface $\sum$ which separates two different regions U^+ and U^- of space–time is called surface shell. We will use index ε ($\varepsilon = \pm$) to denote the physical quantities for each region U^ε. In the general case the hypersurface $\sum$ may be time–like, space–like or light–like. First let us consider the first two cases. Normal Gaussian coordinates (q, x_ε^i) can be introduced in such a manner that we will have $q > 0$ in U^+, $q < 0$ in U^- and $q = 0$ on $\sum$. For these coordinates the metric in the vicinity of $\sum$ takes the form

$$ds_\varepsilon^2 \equiv g_{\mu\nu}^\varepsilon dx_\varepsilon^\mu dx_\varepsilon^\nu = \nu dq^2 + h_{ij}^\varepsilon(q, x_\varepsilon^k) dx_\varepsilon^i \, dx_\varepsilon^j \tag{A1.1}$$

where $\nu = +1$ for the space–like hypersurface and $\nu = -1$ for the time–like hypersurface. Let us denote by n^μ the unit normal vector, in the direction of U^+,

$$n^\mu n_\mu = -\nu. \tag{A1.2}$$

For the Gaussian coordinates we have

$$n^\mu \partial_\mu = \partial_q. \tag{A1.3}$$

In general relativity the intrinsic geometry is well defined on $\sum$ but extrinsic curvature can have a jump. The first condition means that two metrics dh_+^2 and dh_-^2 on $\sum$

$$dh_\varepsilon^2 = h_{ij}^\varepsilon(0, x_\varepsilon^k) dx_\varepsilon^i dx_\varepsilon^j \tag{A1.4}$$

are isometrical. If x^i are some rather arbitrary coordinates on $\sum$, then the corresponding well–defined metric on $\sum$ is written in the form

$$ds^2 = h_{ij}(x^k) dx^i dx^j. \tag{A1.5}$$

The extrinsic curvature $K_{\mu\nu}^{(\varepsilon)}$ of hypersurface $\sum$ is defined as

$$K_{\mu\nu}^{(\varepsilon)} = -p_\mu^{(\varepsilon)\alpha} p_\nu^{(\varepsilon)\beta} \nabla_\alpha^{(\varepsilon)} n_\beta \tag{A1.6}$$

where

$$p_{\mu\nu}^{(\varepsilon)} = g_{\mu\nu}^{(\varepsilon)} - \nu n_\mu n_\nu \tag{A1.7}$$

For the coordinate system (A1.1) we have $K_{ij}^{(\varepsilon)} = -\frac{1}{2}\partial_q h_{ij}^{(\varepsilon)}$.

Let us also introduce the following notation

$$[K_{ij}] = K_{ij}^{(+)} - K_{ij}^{(-)} \tag{A1.8}$$

The Einstein equations

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}\, g_{\mu\nu} R = 8\pi T_{\mu\nu} \tag{A1.9}$$

take the following $(3+1)$ form

$$G^i_j = {}^{(3)}G^i_j - \nu\partial_q(K^i_j - \delta^i_j K) - \nu[K^i_j K - \frac{1}{2}\,\delta^i_k(K_{km}K^{km} + K^2)] = 8\pi T^i_j \qquad (A1.10)$$

$$G^n_j = -\nu(K_{|j} - K^i_{j|i}) = 8\pi T^n_j \qquad (A1.11)$$

$$G^n_n = -\frac{1}{2}\,{}^{(3)}R - \frac{1}{2}\,\nu(K^2 - K_{im}K^{im}) = 8\pi T^n_n \qquad (A1.12)$$

where ${}^{(3)}G^i_j = {}^{(3)}R^i_j - \frac{1}{2}\,\delta^i_j\,{}^{(3)}R$, ${}^{(3)}R^i_j$ is the Ricci tensor for h_{ij} and $(\ )_{|i}$ denotes the covariant derivative for the metric h_{ij}. Integrating (A1.10) over q in the vicinity of $\sum$ we obtain

$$-\nu([K^i_j] - \delta^i_j[K]) = 8\pi S^i_j \qquad (A1.13)$$

where

$$S^i_j = \lim_{\epsilon\to 0}\int_{-\epsilon}^{\epsilon} T^i_j\, dq \qquad (A1.14)$$

Calculating the jumps (A1.11) and using (A1.13) we find

$$S^i_{j|i} = -(T^{(+)n}_{\ \ j} - T^{(-)n}_{\ \ j}) \equiv -[T^n_j]. \qquad (A1.15)$$

Here the notation $A^n \equiv n_\mu A^\mu$ is used.

For the given solutions $g^{(\epsilon)}_{\mu\nu}$ of Einstein equations the system of Eqs. (A1.13) and (A1.14) is the full self–consistent one to attach the different solutions. Another equations are satisfied identically.

The problem of matching the solutions of the Einstein equations at null shells were analyzed in Refs. [10],[19].

APPENDIX 2

The shells with variable parameters in the case of spherically–symmetric space–times.

Let us consider the space with metric

$$ds^2 = f\,dv^2 + 2dvdr + r^2 d\Omega^2 \qquad (A2.1)$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta d\varphi^2$, $f = f(r,v) = -\nabla r\nabla r$ and v is an advanced time coordinate which is normalized at infinity (where $f \to 1$) by the condition $\nabla r\nabla v = 1$. Let $\sum$ be the hypersurface in this space, and the equation

$$\Phi \equiv -r + R(v) = 0 \qquad (A2.2)$$

defines $\sum$. The internal geometry of this hypersurface is described by 3–metric

$$\begin{aligned} dh^2 &= \nu\varepsilon^2(v)dv^2 + R^2(v)d\Omega^2 \\ &= \nu dq^2 + \rho^2(q)d\Omega^2 \end{aligned} \qquad (A2.3)$$

where

$$\begin{aligned} \nu\varepsilon^2(v) &= f(R(v),v) + 2\frac{dR}{dv} \equiv F(v) + 2\frac{dR}{dv} \\ \nu &= sign(\nu\varepsilon^2), dq = \varepsilon(v)dv \\ \rho(q) &= R(v(q)). \end{aligned} \qquad (A2.4)$$

The hypersurface $\sum$ is a space–like one in the case $\nu = +1$ and it is time–like hypersurface for $\nu = -1$. (For discussion of the latter case see also Ref.[17]).

Let us define $U^{(-)}$ as the region of space–time which consists of the points which can be connected to infinity by light rays $v, \theta, \varphi =$ const without intersecting hypersurface $\sum$. All other points form the region $U^{(+)}$.

The vector

$$n^\mu \equiv -\frac{\nu}{\varepsilon}(-1, F + \frac{dR}{dv}, 0, 0) \qquad (A2.5)$$

is the unit normal vector to $\sum$ directed to $U^{(+)}$. To complete this vector up to the tetrad we will also use the other unit normal vectors

$$\begin{aligned} e^\mu_{(1)} &= \frac{1}{\varepsilon}(1, \frac{dR}{dv}, 0, 0) \\ e^\mu_{(2)} &= \frac{1}{R}(0,0,1,0), e^\mu_{(3)} = \frac{1}{R\sin\theta}(0,0,0,1). \end{aligned} \qquad (A2.6)$$

The extrinsic curvature tensor K_{ij} for the hypersurface $\sum$ has the following nonzero components:

$$K^1_1 = \frac{1}{\sigma}[\ddot\rho + \frac{\nu}{2}\frac{df}{dr} + \frac{1}{2\varepsilon^2}\frac{df}{dv}]$$

$$K_2^2 = K_3^3 = \frac{\sigma}{\rho}, \tag{A2.7}$$

where

$$-\sigma = n^\mu \nabla_\mu r = -\frac{\nu}{\varepsilon}\left(F + \frac{dR}{dv}\right) \tag{A2.8}$$

and the point denotes the differentiation in the direction of $e^\mu_{(1)}$:

$$(\ )^{\cdot} \equiv e^\alpha_{(1)} \nabla_\alpha (\ ) = \varepsilon^{-1} \frac{d}{dv}(\ ). \tag{A2.9}$$

Using the Eqs. (A2.8) and (A2.9) it is easy to find that

$$\varepsilon = \sigma + \nu \dot{\rho}. \tag{A2.9'}$$

As a first example let us attach the Schwarzschild $g^{(-)}_{\mu\nu}$ to the de Sitter $g^{(+)}_{\mu\nu}$ metric along the space–like ($\nu = 1$) hypersurface $r = r_0 = \text{const}$ ($\ell < r_0 < 2m$). In this case we have $R(v) = r_0$. The Schwarzschild metric can be rewritten in the form (A2.1) where

$$f_- = -1 + \frac{2m}{r}. \tag{A2.10}$$

It is easy to verify that

$$\varepsilon = \varepsilon_{(-)} \equiv \sqrt{\frac{2m}{r_0} - 1}; \quad \sigma = \sigma_{(-)} = \sqrt{\frac{2m}{r_0} - 1} \tag{A2.11}$$

and

$$dh^2_{(-)} = dq^2 + r_0^2 d\Omega^2. \tag{A2.12}$$

The non–zero components of the extrinsic curvature K^-_{ij} are equal to

$$K^{(-)1}{}_1 = \frac{m}{r_0^2} \frac{1}{\sqrt{\frac{2m}{r_0} - 1}}; \quad K^{(-)2}{}_2 = K^{(-)3}{}_3 = \frac{1}{r_0}\sqrt{\frac{2m}{r_0} - 1} \tag{A2.13}$$

and for the de Sitter metric by an analogous method we find that

$$f_+ = \left(\frac{r}{\ell}\right)^2 - 1; \quad dh^2_{(+)} = dq^2 + r_0^2 d\Omega^2;$$

$$\varepsilon_+ = \sqrt{\left(\frac{r_0}{\ell}\right)^2 - 1} = -\sigma_{(+)},$$

$$K^{(+)1}{}_1 = \frac{r_0}{\ell^2 \sqrt{\left(\frac{r_0}{\ell}\right)^2 - 1}}; \quad K^{(+)2}{}_2 = K^{(+)3}{}_3 = \frac{1}{r_0}\sqrt{\left(\frac{r_0}{\ell}\right)^2 - 1} \tag{A2.14}$$

If we introduce the following notations

$$\varkappa \equiv [K^1_1] = \left(\frac{r_0}{\ell^2 \sqrt{\frac{r_0^2}{\ell^2} - 1}} + \frac{m}{r_0^2 \sqrt{\frac{2M}{r_0} - 1}} \right),$$

$$\lambda \equiv [K^2_2] = -\frac{1}{r_0}\left(\sqrt{\frac{2m}{r_0} - 1} - \sqrt{\left(\frac{r_0}{\ell}\right)^2 - 1} \right), \tag{A2.15}$$

then

$$S^1_1 = \frac{\lambda}{4\pi}; \quad S^2_2 = S^3_3 = \frac{\varkappa + \lambda}{8\pi}. \tag{A2.16}$$

It is easy to verify that $[T^n_i] = 0$ here, and correspondingly Eq. (A1.15) is fulfilled identically.

As the second example we consider the matching of Vaidya ($g^-_{\mu\nu}$) and de Sitter ($g^+_{\mu\nu}$) metrics. When the matter with negative energy density falls into a black hole the Vaidya metric takes the form (A2.1) also and

$$f_- = \frac{2m(v)}{r} - 1 \tag{A2.17}$$

for it. Let us use the hypersurface $\sum$, which is defined by Eq. (A2.2) to attach the Vaidya and de Sitter metrics. Then we have

$$\nu\varepsilon^2_-(v) = \frac{2m(v)}{R(v)} - 1 + 2\frac{dR}{dv},$$

$$\sigma_- = \frac{\nu}{\varepsilon_-}\left(\frac{2m(v)}{R(v)} - 1 + \frac{dR}{dv}\right), \tag{A2.18}$$

and

$$K^{(-)1}{}_1 = \frac{1}{\sigma_-}\left[\ddot{\rho} - \frac{\nu m}{\rho^2} + \frac{\dot{m}}{\varepsilon_- \rho}\right],$$

$$K^{(-)2}{}_2 = K^{(-)3}{}_3 = \frac{\sigma_-}{\rho}. \tag{A2.19}$$

It is easy to find the following expressions for the de Sitter metric

$$f_+ = \left(\frac{r}{\ell}\right)^2 - 1,$$

$$\nu\varepsilon^2_+ = \left(\frac{R(v)}{\ell}\right)^2 - 1 + 2\frac{dR}{dv},$$

$$\sigma_+ = \frac{\nu}{\varepsilon_+}\left[\left(\frac{R(v)}{\ell}\right)^2 - 1 + \frac{dR}{dv}\right]; \tag{A2.20}$$

and

$$K^{(+)1}{}_1 = \frac{1}{\sigma_+}\left[\ddot{\rho} + \frac{\nu\rho}{\ell^2}\right],$$

$$K^{(+)2}{}_2 = K^{(+)3}{}_3 = \frac{\sigma_+}{\rho}. \tag{A2.21}$$

For the jumps of the extrinsic curvature we obtain

$$\varkappa \equiv [K_1^1] = \frac{1}{\sigma_+}\left[\ddot{\rho} + \frac{\nu\rho}{\ell^2}\right] - \frac{1}{\sigma_-}\left[\ddot{\rho} - \frac{\nu m}{\rho^2} + \frac{\dot{m}}{\varepsilon_-\rho}\right]$$

$$\lambda \equiv [K_2^2] = \frac{1}{\rho}[\sigma_+ - \sigma_-], \qquad (A2.22)$$

and

$$S_1^1 = \frac{\lambda}{4\pi}, S_2^2 = S_3^3 = \frac{\varkappa + \lambda}{8\pi}. \qquad (A2.23)$$

The "conservation law" takes the form

$$\dot{\lambda}\rho^2 + (\lambda - \varkappa)\rho\dot{\rho} = -\frac{\nu\dot{m}}{\varepsilon}. \qquad (A2.24)$$

In the simple case, when $\lambda = 0$, Eq. (A2.24) gives

$$\varkappa = \frac{\nu}{\rho\varepsilon}\frac{dm}{d\rho}. \qquad (A2.25)$$

Taking into account (A2.22), we find that $\sigma_- = \sigma_+$ and correspondingly (see $(A2.9')$) $\varepsilon_+ = \varepsilon_-$ or

$$\rho = \ell(\frac{2m}{\ell})^{1/3}. \qquad (A2.26)$$

If we use this equation, then Eq. (A2.25) can be rewritten in the form

$$\varkappa = \frac{3}{2}\frac{\nu\rho}{\ell^2\varepsilon}. \qquad (A2.27)$$

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Figure Captions

Fig. 1 Conformal Penrose diagram for de Sitter space–time.

Fig. 2 Conformal Penrose diagram for an eternal black hole.

Fig. 3 Conformal diagram for the spacetime which is obtained when de Sitter space is attached to the Schwarzschild one on the junction hypersurface of constant radial coordinate ($r_0 = \text{const}$). In this case black hole singularity (at $r = 0$) disappears.

Fig. 4 Topology of the junction hypersurface in the case of space–time which is represented in Fig. 3.

Fig. 5 Conformal diagram for the black hole which was formed by collapsing dust matter.

Fig. 6 Behaviour of an apparent horizon of evaporating black holes.

Fig. 7 Conformal diagrams for evaporating black holes for a variety of parameters characterizing the final state of black holes are schematically represented.

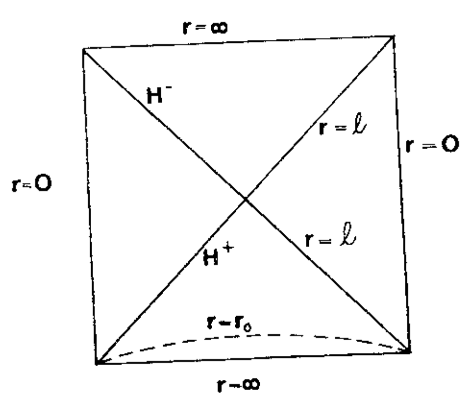
Fig. 8 Conformal diagram for an evaporating black hole in some specific case of stable remnant of black hole.

Fig. 9 Time symmetric diagram of two eternal black holes which are connected by some region of de Sitter space.

Fig. 10 Diagram which shows the development of the interior of the black hole which produces the white hole in a new asymptotically flat Universe.

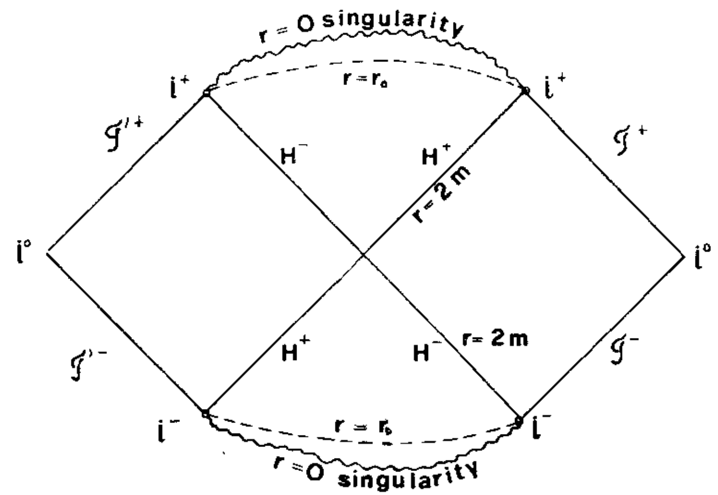
Fig. 11 Conformal diagram in the case when the closed Friedmann Universe is formed in the black hole interior as a result of de Sitter space instability.

Fig. 12 Conformal diagram in the case when the open (or flat) Friedmann Universe is attached to the de Sitter one.



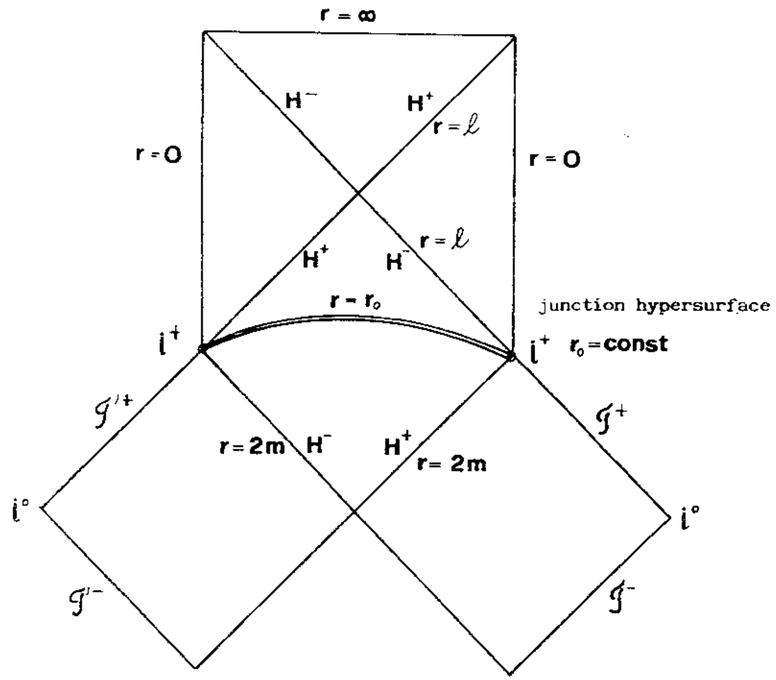
r=∞
H⁻
r=ℓ
r=O
r=O
r=ℓ
H⁺
r=r₀
r=∞

Fig.1



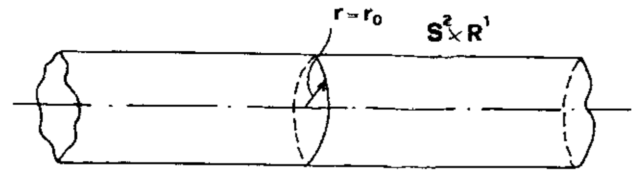
r = O singularity
r = r₀
i⁺
i⁺
𝒥′⁺
𝒥⁺
H⁻
H⁺
r=2m
i°
i°
H⁺
H⁻
r=2m
𝒥′⁻
𝒥⁻
i⁻
i⁻
r = r₀
r=O singularity

Fig.2



r = ∞
r = O
r = O
H⁻
H⁺
r = ℓ
r = ℓ
H⁺
H⁻
r = r₀
junction hypersurface
r₀ = const
i⁺
i⁺
r = 2m
H⁻
H⁺
r = 2m
i°
i°

Fig.3



r = r₀
S² × R¹

Fig.4

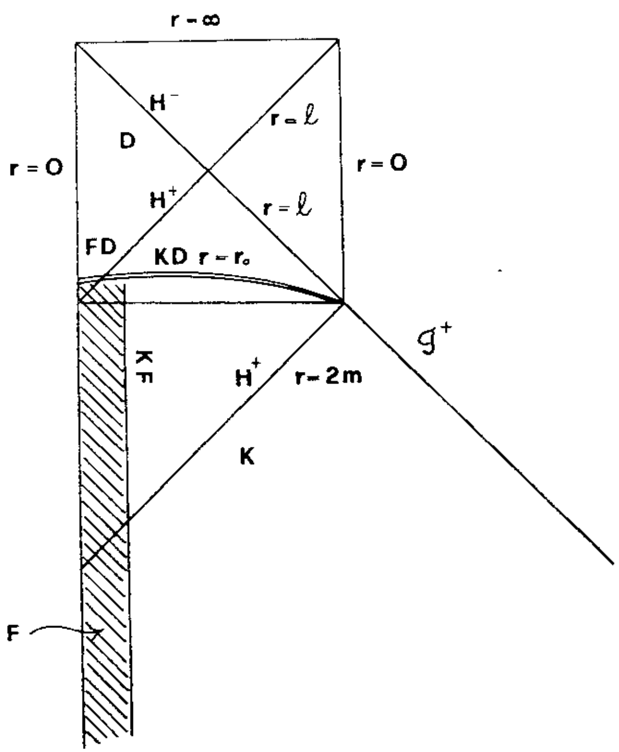

r = ∞
r = O
r = O
H⁻
D
r = ℓ
r = ℓ
H⁺
FD
KD r = r₀
KF
H⁺
r = 2m
K
F
ℐ⁺

Fig.5

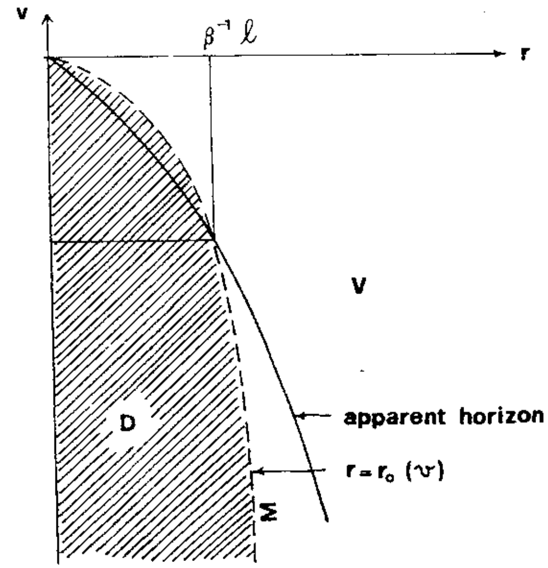

v
r
β⁻¹ ℓ
V
D
apparent horizon
r = r₀ (v)
M

Fig.6

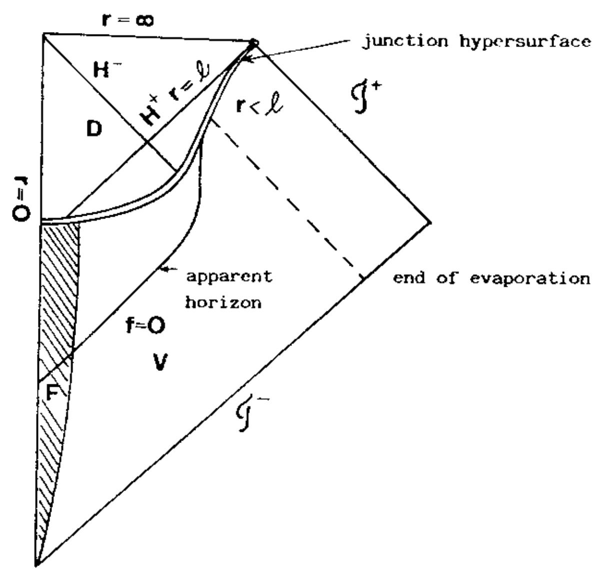


Fig.7a $(\beta > 1)$

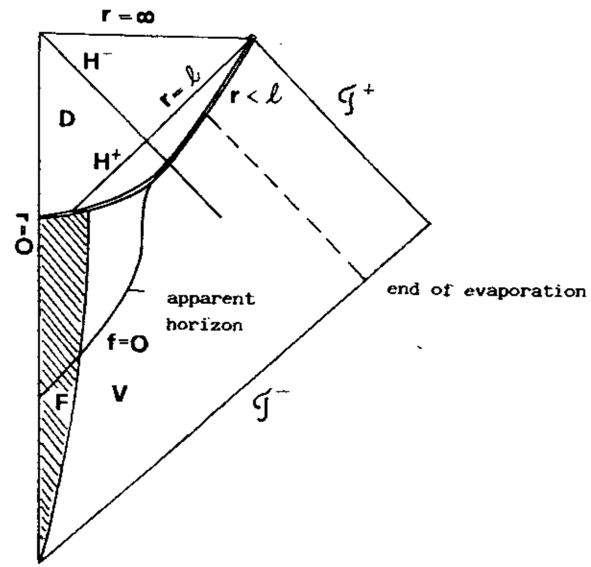


Fig.7b $(\beta < 1)$

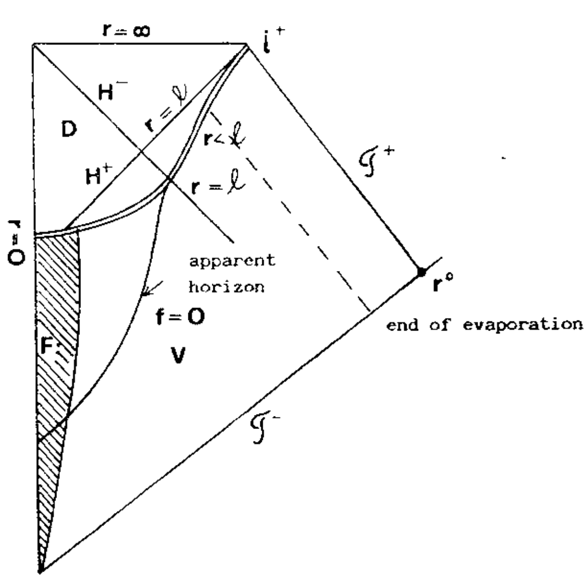


Fig.7c (β = 1)

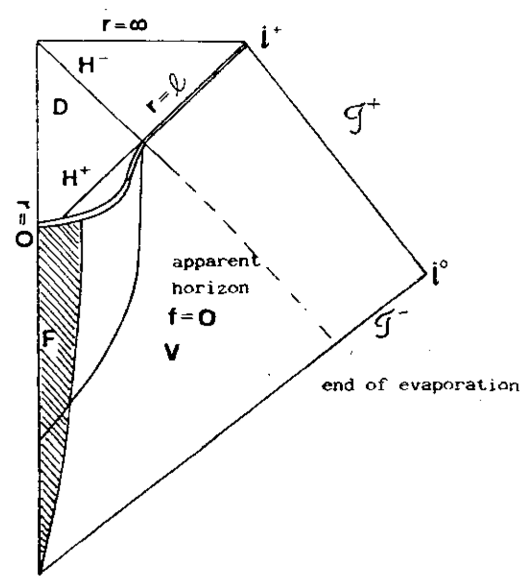


Fig.8

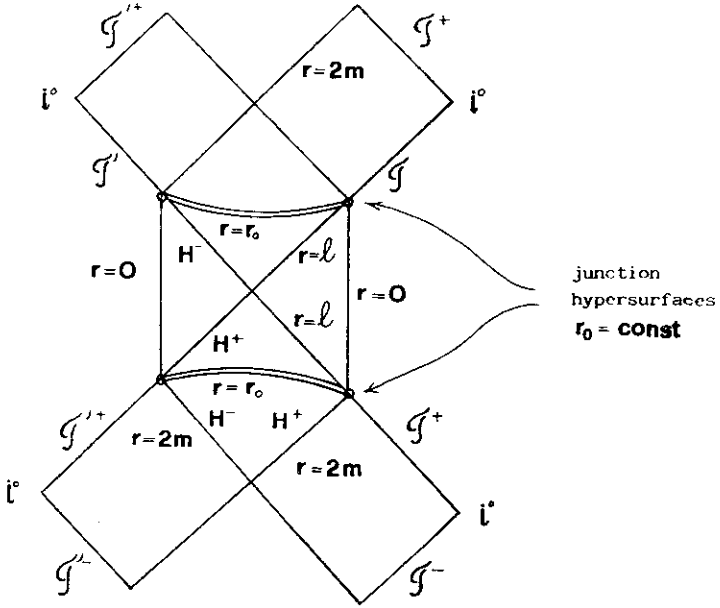

r=2m
r=r_0
H⁻
r=O
r=ℓ
r=O
r=ℓ
H⁺
r = r_0
H⁻
H⁺
r=2m
r=2m
junction
hypersurfaces
r_0 = const

Fig.9

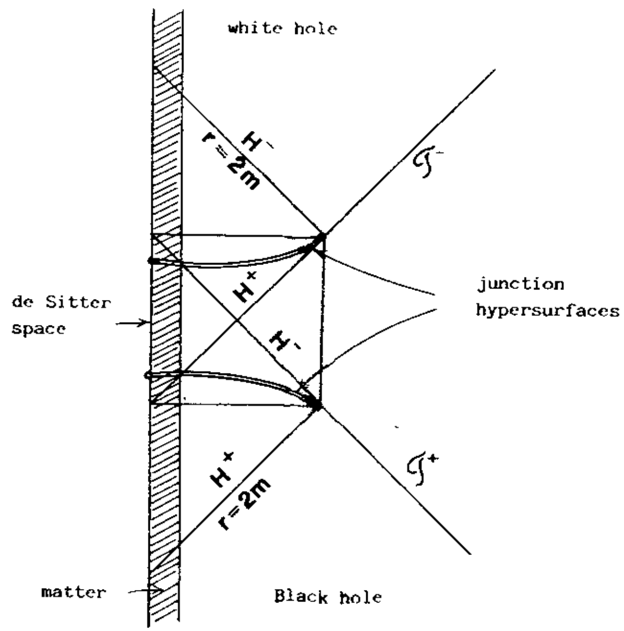
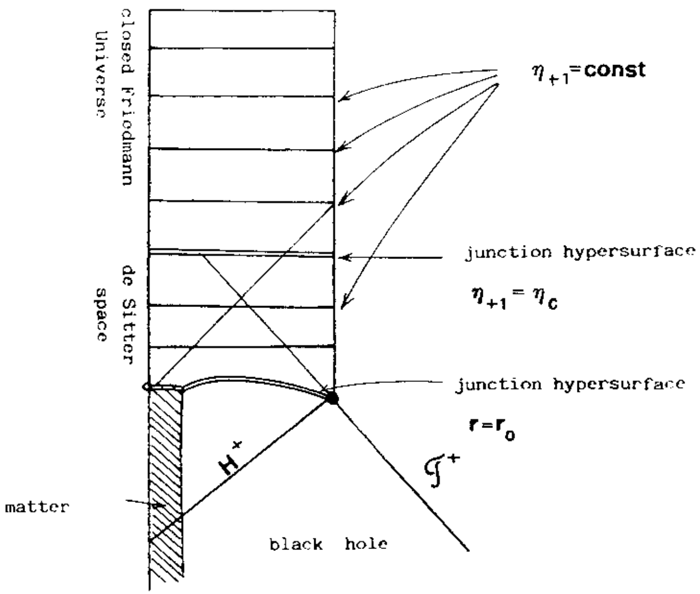
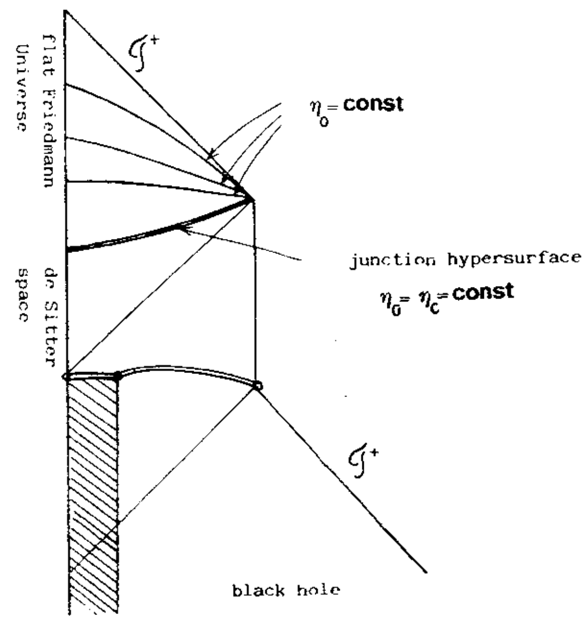

white hole
H⁻
r=2m
junction
hypersurfaces
de Sitter
space
H⁺
H⁻
H⁺
r=2m
matter
Black hole

Fig.10



closed Friedmann Universe
de Sitter space
η_{+1}=const
junction hypersurface
$\eta_{+1} = \eta_c$
junction hypersurface
$r=r_0$
H^+
$\mathcal{J}^+$
matter
black hole

Fig.11



flat Friedmann Universe
de Sitter space
$\mathcal{J}^+$
η_0= const
junction hypersurface
$\eta_0 = \eta_c$= const
$\mathcal{J}^+$
black hole

Fig.12

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